Discriminative Latent-Space Learning for Fine-Grained Object Detection in Remote Sensing Images

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***Abstract*—Fine-grained object detection (FOD) is essential in many remote sensing image interpretation tasks. Existing FOD methods have achieved remarkable progress in modeling discriminative features for FOD in remote sensing images. However, they receive unsatisfactory recognition accuracy due to the curse of dimensionality (CoD) problem. In this article, we propose an orthogonal constraint-based discriminative latent-space learning (DLL) method to address the CoD problem. We first optimize a sparse optimization paradigm with convex relaxation to extract shared features between fine-grained objects into a low-dimensional latent space. Then, we solve the orthogonal space of the latent space for extracting irrelevant features related to common features from input features, i.e., discriminative features. The sparse optimization paradigm reprojects the underlying trends hidden in high-dimensional feature space into a low-dimensional latent space and hence addresses the CoD problem. We use neural parameters with latent space and orthogonal constraints to approximately solve the proposed DLL, which can be efficiently optimized under convex programming tools. We theoretically prove the effectiveness of the proposed DLL. By adding our DLL to the existing deep learning-based object detection method, extensive experiments conducted on two datasets demonstrate that our method achieves superior performance compared with other state-of-the-art methods.**

***Index Terms*—Convex relaxation, curse of dimensionality (CoD), orthogonal space, sparse optimization.**

I. Introduction

FINE-GRAINED object detection in remote sensing images (FOD-RSI) aims to classify fine-grained classes of a superior class and realize precise localization with bounding boxes. Due to its efficiency, FOD-RSI has been widely used in military and civilian applications [1], [2], [3], [4], [5].

As discussed in previous FOD in natural scene image studies, discriminative feature modeling plays a crucial role in fine-grained image recognition [6], [7], [8], [9], [10], [11], [12]. However, in remote sensing images captured from an overhead view on a large scale, objects of interest are not always dominant, resulting in a strong similarity feature space between different fine-grained objects [2]. Consequently, compared with the FOD in natural scene images, accurate FOD-RSI is still challenging due to the inherently subtle interclass differences of objects [2]. In this case, the key issue of FOD-RSI is how to model the discriminative features between different classes of objects under a strong similarity feature space.

Benefiting from the specially designed structures, existing methods have achieved competitive progress in FOD-RSI [1], [2], [3], [5], [6]. However, most neural-network-based methods assume either explicitly or implicitly a certain class of models to learn. In the nonparametric setting, as the number of observations increases, learning algorithms may adapt the complexity of the models [13]. This means that the sample complexity (i.e., the number of observations) to adapt to any particular problem is typically large. In practice, attempting to solve highly complex FOD problems simply using limited data (observations) and preset models leaves a "blind spot"—contiguous regions of feature space without any observations [14]. Such a phenomenon is called the CoD problem, posing several challenges to modeling the discriminative features between fine-grained objects and adversely affecting the model's performance. Fortunately, previous studies have empirically proven that high-dimensional data space often lies in low-dimensional structures instead of uniformly distributed across the ambient space [15]. The subspace methods that recover such low-dimensional structures from high-dimensional feature space can effectively improve the performance of recognition tasks [15].

Considering that the modeling of discriminative features is aggravated in the context of FOD-RSI, it is difficult to directly extend the success of existing subspace learning methods to FOD-RSI. In this case, the key problem of modeling discriminative features is how to design a suitable subspace learning method to model the discriminative underlying trends hidden in a high-dimensional feature space.

Driven by the above analysis, we propose a novel discriminative latent-space learning (DLL) method. As shown in Fig. 1(b), not all local subregions are discriminative features. Existing part-based methods that extract local fixed subregion features may cause suboptimal discriminative feature generation. For example, the red circle denotes a local fixed

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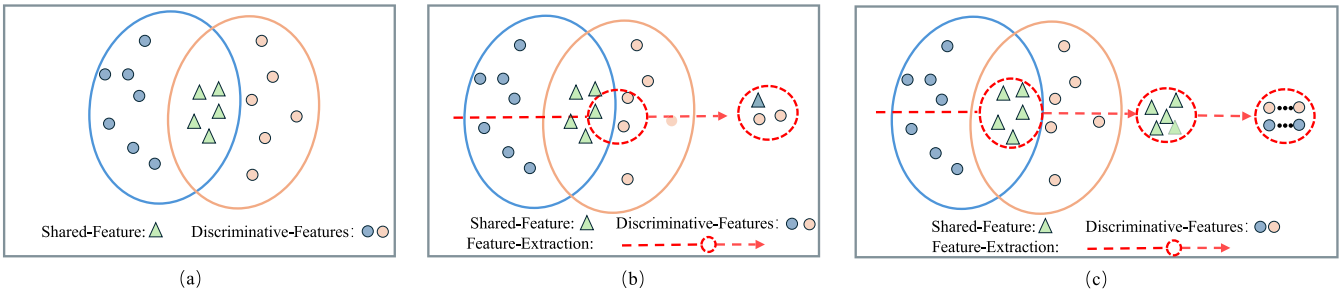


Fig. 1. Comparisons of schematics of feature extraction between existing part-based methods and our proposed method. (a) Two different feature distributions. (b) Schematic of existing part-based methods. (c) Schematic of our proposed method. We use the blue and yellow ellipses to represent two different features, where the cross region denotes similar features.

region in Fig. 1. Simply extracting a fixed region may include shared features and discriminative features and hence cause suboptimal discriminative feature extraction. Different from existing methods, as shown in Fig. 1(c), we first model the shared features and then extract the irrelevant features of the input features to shared feature, called discriminative features. In this way, we can achieve superior discriminative feature extraction. Specifically, our DLL first finds the common underlying feature through latent representation (LR). This representation states that each data point in a union of latent spaces can be efficiently represented as a linear or affine combination of other points [15]. In general, solving such a paradigm is not unique due to the dimension of data points usually being smaller than their number in a latent space. This motivates us to solve a sparse representation optimization. Since solving such a sparse optimization is NP-hard in general, Frobenius relaxation is used to restrict the set of sparse optimization solutions and ensure that its nonzero elements ideally correspond to points from the same latent space. As a result, we can obtain an LR for each data point that models the underlying trends hidden in feature space. After obtaining the common underlying features, we try to extract irrelevant features related to common underlying features from input features, i.e., latent-space discriminative features. According to the existing knowledge of linear algebra, in 2-D space, two vectors parallel and orthogonal to vector A form a set of bases, which can linearly represent any vector. The essence of projecting vector A onto vector B is to preserve parallel parts and eliminate orthogonal parts. This motivates us to think about preserving discriminative parts and eliminating common parts between different fine-grained objects. To achieve this goal, we solve the orthogonal complementary space of the low-dimensional common underlying latent space and map the input features to the orthogonal complementary space. We theoretically prove that our proposed method can effectively extract the latent-space discriminative features. Experimental analyses exhibit that the common underlying trends can reduce the feature distance between different fine-grained objects, and the discriminative latent-space mapping can enlarge the feature distance between different fine-grained objects. This demonstrates the effectiveness of the proposed DLL for modeling discriminative features. We add the proposed DLL to the existing deep learning-based object detection method to validate its effectiveness. Extensive experiments show that the DLL-based method achieves superior performance than existing state-of-the-art methods.

In summary, the major contributions of this article are given as follows.

1) A discriminative feature modeling method specially designed for the CoD problem is proposed for fine-grained object detection (FOD). To the best of the authors' knowledge, the proposed method is the first attempt to address the CoD problem of modeling discriminative features.
2) The proposed method is theoretically and experimentally proven to be effective for modeling discriminative features.
3) We prove a theorem of modeling discriminative features based on orthogonal constraints, providing a theoretical explanation of why the proposed method is effective.

II. Related Works

A. Object Detection

Object detection aims to localize a set of objects and recognize their categories. It has achieved rapid technological evolution and has profoundly impacted the entire computer vision field [16]. Existing object detection methods can be roughly classified into three categories, i.e., two-stage methods [20], [21], [22], [23], one-stage method [21], [22], [23], [24], [25], and anchor-free-based methods [26], [27], [28]. The two-stage-based methods first generate candidate region proposals with an anchor box by using the region proposal network [19], and then, the classification and regression branch takes these proposals for final object box generation.

Although these two-stage-based methods have shown remarkable performance, their demand for extensive computational resources limits the applications in the real world. To tackle the above limitation, one-stage-based method is proposed. It directly utilizes classification and regression branches without FPN and thus achieves more efficiency than the two-stage-based methods. Recently, due to the hyperparameters related to anchor boxes being sensitive to the final detection performance, anchor-free-based methods are proposed to directly predict the coordinates of a bounding box based on the deep learning features.

B. Fine-Grained Object Recognition

Based on object detection studies, fine-grained object recognition aiming to classify subordinate categories has attracted the attention of many researchers. Benefiting from the development of deep learning, fine-grained object recognition has

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witnessed remarkable progress [8]. As discussed in [6], existing methods can be organized into three paradigms, i.e., localization-based methods, feature-encoding-based methods, and external information-based methods. The localization-based methods try to capture the semantic parts of fine-grained objects with distinguishing information to classify fine-grained objects. For example, Yang et al. [10] proposed a navigator–teacher–scrutinizer network with a self-supervision mechanism focused on distinguishing part features of fine-grained objects. The navigator–teacher–scrutinizer network evaluates the part feature extracted by the navigator, and the scrutinizer scrutinizes this part feature for classification. Part-based CNN [8] tried to locate the discriminative parts of fine-grained objects. It constructs a subnetwork that classifies the semantic representation of these parts for the final classification. Different from conventional methods with explicit network designs for part detection, Han et al. [8] designed a mutual channel loss to learn part localized discriminative features.

The differences between fine-grained objects are strongly subtle. Capturing discriminative information by conventional layers limits the representation capacity of the discriminative features of a model. The feature-encoding-based methods were proposed to extract high-order feature interactions. Lin et al. [29] proposed bilinear CNN models using the covariance matrix-based representation method to encode second-order statistics of convolutional activations. The proposed models improved the representation capacity of discriminative features. Min et al. [30] integrated three bilinear representations in terms of square root, low-rank, and sparsity to normalize matrix power for stabilizing the second-order information in bilinear features.

The external information-based methods leverage external information to assist fine-grained recognition. Reed et al. [31] designed a multimodal interaction method that jointly embeds images and fine-grained visual descriptions of text for better distinguishing fine-grained categories. Zhang et al. [32] supervised the separation of useful and irrelevant noisy web images by utilizing the prediction scores in different training epochs.

C. Subspace Representation

Although the above fine-grained method has been experimentally proven effective, existing methods ignore the CoD problem of modeling discriminative features. Fortunately, existing works like [15] have demonstrated the effectiveness of subspace methods for alleviating the CoD problem. Subspace clustering aims to categorize input data based on their underlying subspaces. Subspace-based methods state that objects from different categories can be accurately represented using a low-dimensional subspace of the high-dimensional ambient space [15]. Due to its effectiveness in recognition, it has been successfully used in numerous applications.

Existing subspace clustering methods related to this article can be divided into three categories, i.e., algebraic-geometric methods, statistical methods, and self-representation-based methods. The algebraic-geometric methods, such as [33] and [34], use a polynomial to represent the subspaces whose derivatives at a point give the normal vector to the subspace passing through the point. Random sample consensus [35] is a statistical method used for fitting a subspace to a randomly chosen subset of points. The process is repeated to find the next subspace by removing the inliers until the number of inliers is large enough. The self-representation-based methods find the representations of a data point in terms of other points. Among the infinitely many possible representations of a data point in terms of other points, sparse representation corresponds to selecting a few points from the same subspace [15]. Different from constructing the sparsest representation for each data point, low-rank representation used the structured sparsity loss to jointly find the lowest rank representation of all data [36].

Based on the subspace representation, LR methods have been developed to learn the projection of data. Zhang et al. [36] integrated underlying complementary information from multiple views into a latent space for comprehensively depicting data. Patel et al. [37] learned the projection of data and founded the sparse coefficients for representing data in latent space. Latent low-rank representation [39] extracts jointly observed and hidden data for constructing a dictionary.

As discussed in the previous works [36], [37], [38], LR simultaneously explores the relationships between data points and achieves dimensional reduction. This representation reveals the common underlying structure shared by different data. It has been demonstrated that LR is effective for various clustering tasks [36]. Inspired by this, we use the LR to generate discriminative features that counteract the common underlying features between fine-grained objects. In this way, we can obtain irrelevant features related to common underlying features from input features.

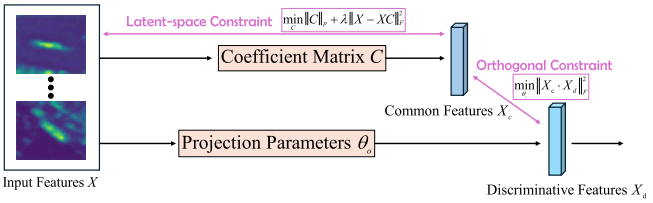


Fig. 2. Overview of the proposed DLL. The coefficient matrix and projection parameters are employed to extract common features and discriminative features, respectively, which are updated under the latent space and orthogonal constraints.

III. Discriminative Latent-Space Learning

A. Problem Formulation

Given a group of images $\mathcal{I} = \{\mathcal{I}_1, \mathcal{I}_2, \ldots, \mathcal{I}_n\}$ with fine-grained objects, our objective is to learn a projection parameter that produces discriminative features f_d

$$f_d = p(\mathcal{I}; \boldsymbol{\theta}) \tag{1}$$

where θ is the projection parameters to be learned. In this article, we propose a DLL method to map the input features into a latent space for dimension reduction and modeling discriminative features between different fine-grained objects. DLL consists of an LR module and a DOD module. The overview diagram of the proposed DLL is shown in Fig. 2. We first use the coefficient matrix with latent-space constraint and projection parameters with orthogonal constraint to project the input features to latent space and discriminative space,

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respectively. The latent-space constraint constrains the input features to the latent subspace. The orthogonal constraint ensures the extraction of discriminative features. The details of these modules are given as follows.

B. Latent Representation

As discussed in previous work [36], the LR depicts the essence of the data and reveals the common underlying structure shared by different data. The LR model is employed to extract the common underlying features shared between different fine-grained objects while achieving dimension reduction. To this end, LR is performed to encode such common features into a latent space. Specifically, for a given feature data $X = \{x_1, x_2, \ldots, x_n\} \in \mathbb{R}^{d\times n}$ drawn from a union of linear subspaces $S = \{s_1, s_2, \ldots, s_k\}$, we try to mine the LR while preserving the original feature information as much as possible.

For LR, we use a self-representation to express each data point as a linear combination of other points

$$x_i = Xc_i \tag{2}$$

where $\mathrm{C} = [c_1, c_2, \ldots, c_n]^T \in \mathbb{R}^{n\times n}$ denotes the coefficient matrix. However, the solution of C is not unique in general due to the dimension of data points in a subspace is often smaller than its number [15]. This motivates us to restrict the set of solutions of C. The following theorem [40] has proven that by minimizing certain norms of c_i, the solution of (1) is block diagonal, corresponding to points in the same subspaces.

Theorem 1: Data $X \in \mathbb{R}^{d\times n}$ drawn from a union of independent linear subspaces $\{S_i\}_{i=1}^k$. Assume $X = [x_1, x_2, \ldots, x_n]\Gamma$, where $\Gamma \in \mathbb{R}^{n\times n}$ denotes an permutation matrix. The solution to (3) $C^* = [c_1^*, c_2^*, \ldots, c_n^*]\Gamma$ is block diagonal.

Proof: c_i^* is a vector of minimum ℓ_p norm that satisfy $x_i = X_{-i}c_i^*$ if c_i^* is a solution in (3). Then c_i^* is decomposed as $c_i^* = c_i + h_i$, where c_i and h_i are the coefficient corresponding to the true space of x_i and other space, respectively. Thus, $X_{-i}c_i^* = X_{-i}(c_i + h_i) = X_{-i}c_i + X_{-i}h_i$. Then, we have $X_{-i}h_i = 0$ since $x_i \in S_i$ and $X_{-i} \in S_i$ and the independence assumption, $X_{-i}h_i \notin S_i$. Consider that c_i and h_i have support on disjoint subset of indices, and assume $h_i \not\simeq 0$, we have $\|c_i\|_p < \|c_i + h_i\|_p = \|c_i^*\|_p$, contradicting the optimality of c_i^* and hence $h_i = 0$.□

C. Discriminative Orthogonal Decoupling

Based on the LR module, we can obtain the common features. Then, we try to extract irrelevant features related to common underlying features from the input features. The DOD module is proposed for modeling discriminative features. Specifically, given the input features and their LR, we want to extract irrelevant features related to common underlying features from the input features. Mathematically, let A be the input features and C be the common features; we have the following definition.

Definition 1: Irrelevant features of A–C are formulated as $A \not\leftarrow C$. Let $\rightarrow$ be a vector projection mapping

$$A \not\leftarrow C = A - A \rightarrow C.$$

According to the formula of vector projection and the definition of linear algebra, two vectors $\mathcal{A}$ and $\mathcal{B}$ are orthogonal if $\mathcal{A} \times \mathcal{B} = 0$; we have the following theorem.

Theorem 2: Solving $A \not\leftarrow C$ is equivalent to the projection of A onto the orthogonal subspace of C.

Proof: Based on the Definition 1, we have

$$(A \not\leftarrow C) \cdot C = (A - A \rightarrow C) \cdot C.$$

According to the standard vector projection formula

$$a \cdot b = \frac{a \cdot b}{|b|^2} b$$

we have

$$\begin{aligned}(A - A \rightarrow C) \cdot C &= \left(A - \frac{A \cdot C}{|C|^2} C\right) \cdot C \\ &= A \cdot C - \frac{A \cdot C}{|C|^2} C \cdot C \\ &= A \cdot C - \frac{A \cdot C}{|C| \cdot |C|} C \cdot C \\ &= 0.\end{aligned}$$

This demonstrates that $A \not\leftarrow C$ and C are orthogonal. So, the projection of A onto the orthogonal subspace of C is equivalent to solving $A \not\leftarrow C$, which concludes the proof. □

IV. DLL-Based Method

A. Overview

Based on the existing object detection method OR-CNN [42] and DLL, the overview of the DLL-based method is shown in Fig. 3. Specifically, the input image is fed into the ResNet [43] for feature map extraction. Then, the proposed LR and DOD modules are used to extract discriminative features. Based on the RoI-extractor and discriminative feature, the RoI-feature is obtained. The RoI-align module is used to align the RoI-feature for dimension alignment. Finally, the aligned feature is fed to the classification and regression layer for result generation.

B. DLL-Based Method

Considering the proposed method is a deep learning-based method, the coefficient matrix and projection parameters are represented with fully connected linear layers. Based on the classification and regression tasks of conventional object detection, we have the following optimization function:

$$\begin{aligned}\min_{\Theta,C} \frac{1}{2}\|X - \Theta X_r\|_F^2 + \lambda_1 \|C\|_F^2 \\ + \frac{\lambda_2}{2}\|X_r - X_r C\|_F^2 + \frac{\lambda_3}{2}\|X_o \cdot \Theta X_r\|_F^2 \\ + \ell_{\mathrm{ce}}(\Theta X, Y) + \ell_{\mathrm{reg}}(\Theta X, Y)\end{aligned} \tag{3}$$

where Θ denotes the network parameters, $X_o = X_r C$, $\ell_{\mathrm{cls}}(\cdot,\cdot)$ denotes the cross-entropy function for measuring the loss of the classification task, and $\ell_{\mathrm{reg}}(\cdot,\cdot)$ is the smoothL1 function for calculating the loss of bounding-box regression, which are represented as

$$\ell_{\mathrm{clas}} = \sum_{j=0}^{K} y_{ji} \log\left(f\left(x_{ji}\right)\right) \tag{4}$$

$$\ell_{\mathrm{reg}} = \sum_{i\in\hat{x}} \mathrm{smooth}_{L1}\left(\hat{x}_i - l_i\right) \tag{5}$$

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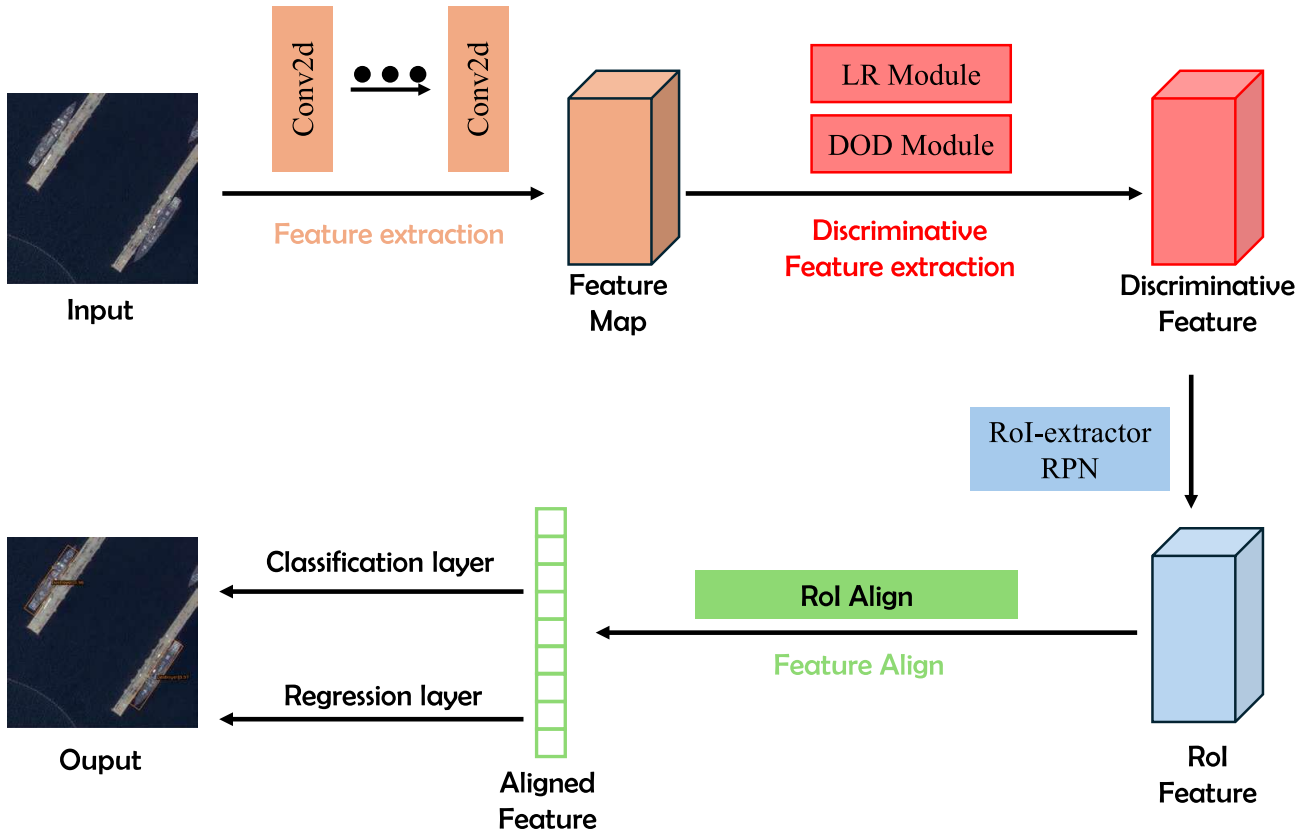


Fig. 3. Overview of the DLL-based method, which is based on the oriented R-CNN (OR-CNN) [42]. The feature extraction consists of a series of convolutional modules that extract semantic feature maps of input images. Then, our proposed method is applied to encode feature maps for discriminative feature generation. The RoI-extractor with RPN is used to produce object candidate boxes and their corresponding object features in input images, which is called the RoI-feature. The RoI-align resizes all RoI-features to the same size, avoiding quantization errors and generating aligned features. The classification and regression layers are used to locate and classify objects.

where y_{ji} is the predicted probability for the ith sample belonging to class j, $\hat{x}$ denotes the output of the network, l_i represents the label of bounding box, $f(\cdot)$ is the activation function, and the $\text{smooth}_{L1}()$ is smoothL1 loss, which are defined as

$$f(a_i) = \frac{e^{a_i}}{\sum_{i=0}^{N} e^{a_i}}. \tag{6}$$

$$\text{smooth}_{L1}(a) = \begin{cases} 0.5a^2, & \text{if } |a| < 1 \\ |a| - 0.5, & \text{otherwise.} \end{cases} \tag{7}$$

Accordingly, the derivative of $f(a_i)$ can be calculated as

$$\frac{\partial f(a_j)}{\partial a_k} = \begin{cases} -f(a_i) \cdot f(a_k), & k \neq j \\ f(a_i) - f(a_i) f(a_k), & k = j. \end{cases} \tag{8}$$

We use the Frobenius norm to ensure that its nonzero elements ideally correspond to points from the same latent space, which is proven by [43]. Then, the objective function in (7) can be solved as follows.

1) *Updata* Θ: Let α and β be the tradeoff parameter, we have

$$\begin{aligned} \frac{\partial \mathcal{L}_\Theta}{\partial \Theta} = {} & X_r^T X_r \Theta_r - X_r^T X_r + \alpha \Theta_r \\ & + \lambda_3 \|X_o \cdot \Theta_o X_o\|_F^2 + \beta \|\Theta_o\|_F^2 \\ & + X(f(\Theta X) - Y) + \upsilon(\Theta) \end{aligned} \tag{9}$$

where $\upsilon(\Theta)$ is defined as

$$\upsilon(\Theta) = \begin{cases} \Theta^2 X, & \text{if } \Theta X < 1 \\ \pm \Theta, & \text{otherwise.} \end{cases} \tag{10}$$

2) Updata C: Based on the derivative of $\mathcal{L}_C$ in (1), we have

$$\mathcal{L}_C = \lambda_1 \|C\|_F^2 + \frac{\lambda_2}{2} \|X_c - X_c C\|_F^2 \tag{11}$$

$$\frac{\partial \mathcal{L}_C}{\partial C} = \lambda_2 X_c^T X_c C - \lambda_2 X_c^T X_c + \lambda_1 C. \tag{12}$$

The fact that C can be considered the parameters of an additional network layer. Thus, we use the gradient descent to achieve an approximate solution. The optimization procedure is summarized in Algorithm 1.

Note that our proposed method is based on conventional object detection and thus causes additional computational costs. Based on (3), we can observe that the first to fourth items of (3) cause additional computational costs. Given $\Theta \in \mathbb{R}^{d \times k}$, $X_r \in \mathbb{R}^{k \times n}$, and $X_o \in \mathbb{R}^{k \times n}$, the complexities of forward calculation are $O(dkn + knn + rn + n^2)$. Considering the gradient calculation, for updating Θ, the complexities are $O(drn + rn^2)$. For updating C, the complexities are $O(knn + kn^2)$. Therefore, based on the maximal iteration number T, the overall complexities of backward propagation are $O(dkn + knn + dknn + kn^2 + n^2)$.

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Algorithm 1 Update Network With the SGD Algorithm

Input: Data $X \subset \{x_1, \ldots, x_n\}$, hyperparameter $\{\lambda, \alpha, \beta\}$, label $Y \subset \{y_1, \ldots, y_n\}$ network parameter Θ, and C learning rate η, and maximal iteration number$T$

Initialization Initialize randomly, Θ, C and $t = 1$

select randomly $x \subset \mathbf{X}$ and $y \subset \mathbf{Y}$

While $t < T$ **do**

Update Θ by (9)

Update C by (12)

$t = t + 1$

end

Output Θ and C

V. Experiment

A. Datasets

We evaluate the proposed method on two challenging datasets, including Fair1M [4] and ShipRSImageNet [44], which are widely used in the FOD-RSI task.

1) Fair1M: Fair1M is a large benchmark dataset of FOD-RSI, which contains 24 625 images, including 16 488 images training set and 8137 images test set. All the images are collected across various countries and with a resolution of 0.3–0.8 m. It contains more than 1 million instances annotated with five coarse categories and 37 fine-grained categories: ship (Engineering Ship, Liquid Cargo Ship, Dry Cargo Ship, Warship, Passenger Ship, Motorboat, Fishing Ship, Tugboat, and Other Ship), vehicle (Small Car, Bus, Cargo Truck, Dump Truck, Van, Trailer, Tractor, Excavator Truck Tractor, and Other Vehicle), airplane (Boeing 737, Boeing 747, Boeing 777, Boeing 787, ARJ21, C919, A220, A321, A330, A350, and Other Airplane), court (Baseball Field, Basketball Court, Football Field, and Tennis Court), and road (Roundabout, Intersection, and Bridge).

2) ShipRSImageNet: ShipRSImageNet is a large-scale fine-grained dataset for ship detection in remote sensing images. It contains over 3435 images with 17.573 ship instances in the dock and 49 categories of ship, including AOE, Arleigh Burke, Asagiri, Atago, Austin, Barge, Cargo, Commander, ContainerShip, Enterprise, EPF, Ferry, Fishing Vessel, Hatsuyuki, Hovercraft, Hyuga, LHA, LSD 41, Masyuu AS, Medical Ship, Midway, Motorboat, Nimitz, OilTanker, Osumi, Other Aircraft Carrier, Other Auxiliary Ship, Other Destroyer, Other Frigate, Other Landing, Other Merchant, Other Ship, Other Warship, Patrol, Perry FF, RORO, Sailboat, Sanantonio AS, Submarine, Test Ship, Ticonderoga, Training Ship, Tugboat, Wasp, Yacht, YuDao, YuDeng, YuTing, and YuZhan.

B. Implementation Details

A PC with Intel[1] Xeon[1] Gold 6226R CPU at 2.90 GHz, 256-GB, 3000-MHz RAM, NVIDIA RTX A6000 (with 48-GB memory), and NVIDIA GeForce RTX 4090 (with 24-GB memory) are used for both training and evaluating. The proposed method and comparison methods are implemented with the PyTorch platform. During training, we use the ResNet-50 [43] as the backbone for the proposed method and comparison unless otherwise specified. The mini-batch size is set to 4. SGD [42] optimizer is used to train our proposed method, and the comparison methods and its hyperparameters are set with initial learning rate lr = 0.005, moment = 0.9, eps = 1e-4, and weight decay = $5e^{-4}$. We crop the original images into 1024×1024 pixels and 800×800 pixels for the Fair1m dataset and ShipRSImageNet dataset, respectively. Random flipping is applied during training for data augmentation. OR-CNN [42] is used as the baseline. Mean average precision (mAP) is used as the evaluation metric that is commonly used in the FOD task.

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C. Comparisons With the State-of-the-Art Methods

To evaluate the proposed method, we compared it with existing state-of-the-art methods, including OR-CNN [42], PCLDet [1], LSKNet [46], FCOS [47] RoI-Transformer [48], Faster R-CNN [19], RetinaNet [23], R3Net [49], S2aNet [50], Gliding Vertex [51], CSL [52], ReDet [53], ATSS [54], LoRA-det [55], TIOE-det [56], and Manba-YOLO [57]. We use the public library MMrotate [58] to train Rotate Faster R-CNN and Rotate RetinaNet for oriented object detection.

1) Qualitative Comparison: Fig. 4 displays the detection results of the proposed method on the Fair1M dataset. As shown, the proposed method can effectively detect most objects under different challenging scenes intuitively, e.g., city scene (first column), sea scene (second column), small object scene (third column), and complex background (fourth column). This demonstrates the effectiveness of the proposed method.

Fig. 5 illustrates the visual comparison with other state-of-the-art methods on the ShipRSImagesNet dataset. As shown, we select four scenes for visualization to verify the robustness of the proposed method, i.e., building interference (first row), equipment obstruction (second row), highly similar dense objects (third row), and low contrast ratio (fourth row). It can be seen that the proposed method performs better in terms of the detection and recognition of fine-grained objects compared with other state-of-the-art methods. For the building interference scene, our method exhibits a better ability for object localization and recognition. Compared with the ReDet, our method accurately localizes and recognizes the dock and cruiser object, while the ReDet fails to achieve this. The R3Net and ATSS localize the dock object with a mismatched bounding box.

For the obstruction scene, the ReDet cannot localize and recognize the objects. Compared with the R3Net and ATSS, our method achieves higher confidence and boundary localization.

For the scene of highly similar dense objects, the R3Net and ReDet fail to accurately localize the five objects with a finely matched bounding box. The ATSS is unable to detect the Frigate ship. Compared with these methods, the visual result of our proposed method is closer to the GT.

For the scene with a low contrast ratio, our method can accurately recognize the two objects with correct localization. However, the R3Net cannot recognize the two objects, and the ReDet and ATSS fail to accurately recognize the Other Ship object.

2) Quantitative Comparison: Table I reports the compari son results on the ShipRSImageNet dataset. It can be found that our proposed method achieves the second-best

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Fig. 4. Detection results of our proposed method on the Fairm1M dataset. Since the Fair1M dataset has not made the ground truth of the test set publicly available, we only illustrate the results of our results.

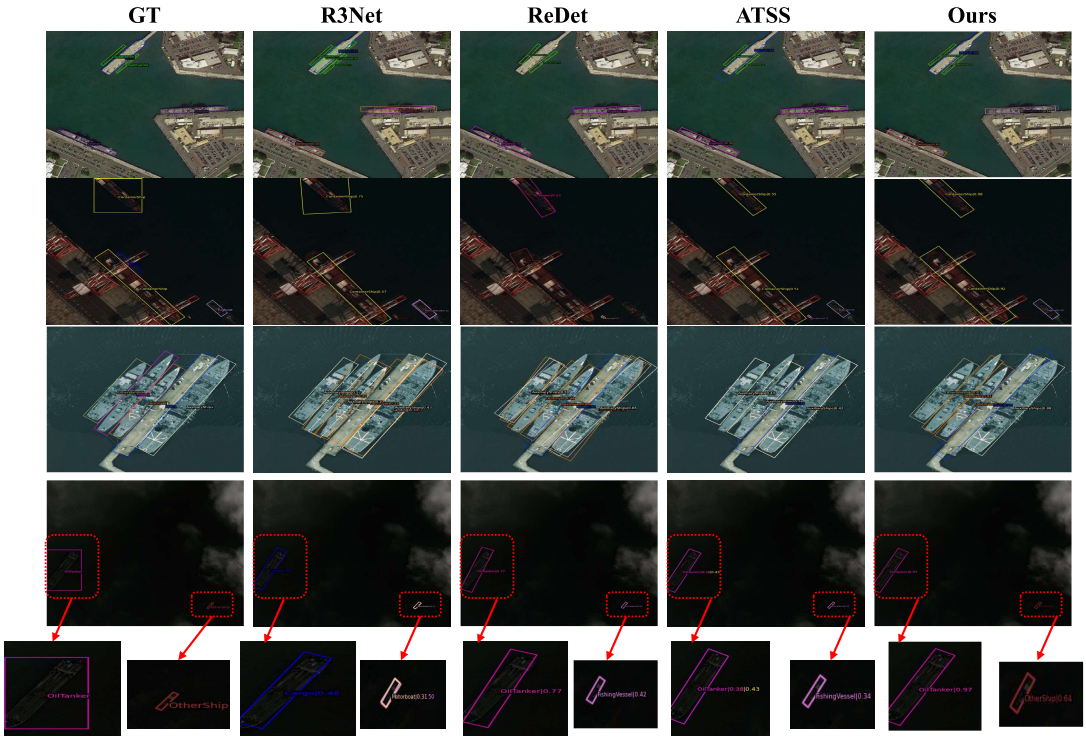


Fig. 5. Qualitative comparison of our proposed method with other state-of-the-art methods, R3Net [49], ReDet [53], and RoI-Transformer (RoI-Trans) [48].

TABLE I

QUANTITATIVE RESULTS IN TERMS OF MAP ON THE SHIPRSIMAGENET DATASET. THE OR-CNN DENOTES ORIENTED R-CNN [42]. THE BEST RESULT IS GIVEN IN BOLD. MSTRAIN REPRESENTS MULTISCALE TRAINING. WE USE RED, BLUE, AND GREEN TO HIGHLIGHT THE BEST, SECOND-BEST, AND THIRD-BEST PERFORMANCE, RESPECTIVELY

Methods	LoRA-Det	TIOE-Det	CSL	RetinaNet	ATSS	ReDet	R3Net	Faster R-CNN	FCOS	S2aNet	LSKNet	Gliding Vertex	RoI-Trans	OR-CNN	PCLDet	DLL(ours)
	50.8	52.8	14.8	17.8	27.7	29.2	30.6	39.0	40.3	43.4	43.7	44.6	47.3	**55.1**	**56.1**	**55.8**
MSTrain	54.6	56.4	18.8	22.3	32.2	37.7	47.0	40.1	42.3	46.2	50.5	43.4	49.1	**57.5**	**58.4**	**58.6**

performance, 0.3 mAP lower than the best method, i.e., PCLDet. Under multiscale training, our proposed method achieves the best performance, with 0.2-mAP improvement compared with the PCLDet in terms of mAP.

Table II shows the comparison results of our proposed method with other state-of-the-art methods on the Fair1M dataset. We can see that our proposed method achieves superior performance compared with other methods.

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TABLE II

QUANTITATIVE RESULTS IN TERMS OF MAP ON THE FAIR1M DATASET. WE USE RED, BLUE, AND GREEN TO HIGHLIGHT THE BEST, SECOND-BEST, AND THIRD-BEST PERFORMANCE, RESPECTIVELY

Methods		TIOE-Det	Manba-YOLO	PCLDet	LSKNet	FCOS	RoI-Trans	FasterR-CNN	RetinaNet	R3Net	S2aNet	Gliding Vertex	CSL	ReDet	ATSS	OR-CNN	DLL(ours)
Airplane	Boeing737	37.62	-	23.63	1.14	28.69	31.49	29.37	32.73	32.91	33.34	30.49	33.44	29.10	34.32	33.12	33.43
	Boeing747	86.71	-	81.50	7.01	79.41	81.83	83.31	76.34	78.71	78.81	81.11	75.83	73.91	71.15	84.75	82.65
	Boeing777	11.06	-	13.37	1.34	15.38	9.15	12.85	12.07	8.56	14.21	9.54	11.67	18.95	9.45	16.52	11.22
	Boeing787	43.32	-	48.62	1.74	35.71	45.79	43.33	41.67	36.74	33.75	42.38	37.08	29.67	35.35	46.41	49.77
	C919	0	-	12.31	0	0.79	9.56	7.92	0	0	0	5.56	0	2.12	0.12	14.87	12.03
	A220	48.75	-	40.18	3.25	39.98	45.22	45.69	36.88	40.36	44.98	44.64	41.00	41.92	39.80	45.07	46.35
	A321	68.49	-	59.96	4.27	60.16	63.71	60.85	62.18	65.9	64.34	60.00	60.57	60.56	60.69	66.32	68.48
	A330	72.51	-	56.84	9.57	57.07	58.90	67.38	51.87	50.01	52.97	65.46	42.40	57.47	48.37	65.59	67.96
	A350	78.19	-	61.35	7.01	71.00	62.36	64.04	64.03	60.30	64.61	63.59	57.13	60.82	61.21	69.24	71.14
	ARJ21	8.62	-	28.33	0.02	20.78	27.68	25.18	1.78	0.80	4.41	29.59	1.333	41.23	0.10	18.79	24.29
Ship	Passenger Ship	3.73	-	10.06	8.05	3.69	8.74	7.00	1.05	2.11	5.51	9.78	0.31	4.98	4.13	12.82	14.78
	Morotboat	58.45	-	51.50	46.77	39.65	50.04	44.84	9.88	24.92	40.79	47.92	10.69	38.02	34.54	54.20	58.12
	Fishing Boat	5.12	-	6.18	3.46	4.06	4.79	4.41	0.48	4.79	5.89	4.08	37.27	3.71	5.20	8.22	9.40
	Tugboat	30.51	-	25.29	11.53	26.39	37.11	32.21	8.17	19.00	23.16	34.80	10.19	8.82	21.19	37.99	35.42
	Engineering Ship	10.38	-	9.29	2.87	10.56	8.15	9.12	8.67	6.80	6.54	10.01	8.74	5.27	7.78	9.69	12.56
	Liquid Cargo Ship	5.56	-	14.14	14.36	9.46	14.19	14.83	0.73	12.16	16.35	12.97	0.28	9.27	11.09	19.06	21.36
	Dry Cargo SHip	18.71	-	29.29	28.58	24.56	29.03	25.61	10.40	30.43	29.71	26.43	9.35	26.77	27.10	36.65	38.01
	WarShip	2.52	-	11.43	17.08	7.45	8.22	9.01	0.01	7.39	8.80	9.26	0.02	3.27	6.17	28.06	24.15
Vehicle	Small Car	65.89	-	47.73	24.12	27.97	54.16	48.87	31.08	41.11	55.45	49.76	31.07	46.49	48.80	54.58	55.66
	Bus	4.73	-	20.85	3.47	9.56	15.25	14.47	0.33	2.22	6.14	18.71	0.08	3.50	3.83	25.43	25.60
	Cargo Truck	36.29	-	31.39	21.45	21.33	32.81	30.26	5.86	20.42	27.60	31.72	6.27	21.77	24.61	37.43	38.92
	Dump Truck	41.31	-	34.16	15.10	13.36	35.12	30.12	5.12	9.32	27.25	34.73	5.07	22.31	17.32	39.65	44.27
	Van	65.89	-	42.70	36.93	24.76	49.46	41.64	20.69	31.69	47.48	43.45	20.55	39.52	40.03	49.33	50.59
	Trailer	0.53	-	5.31	1.25	1.80	4.89	4.00	0	0.38	1.49	6.81	0	0	0.02	8.30	10.66
	Tractor	0.18	-	0	0.10	0	0	1.50	0	0	0	0	0	0	0	5.64	4.46
	Excavator	9.83	-	11.15	4.07	6.24	7.82	7.49	0	0	0.757	8.85	0	0	4.72	10.90	16.35
	Truck Tractor	0.10	-	0	0.41	0	0	0	0	0	0.15	0	0	0	26.73	0.01	0.28
Court	Basketball Court	50.23	-	36.02	23.91	30.73	40.02	39.73	13.46	24.03	25.47	38.57	13.91	10.84	26.73	43.16	46.90
	Tennis Court	80.23	-	78.0	56.84	76.40	77.96	77.59	75.83	75.43	75.81	77.11	72.63	68.28	76.52	78.88	81.21
	Football Field	60.70	-	53.5	38.07	44.87	54.14	47.70	50.89	49.13	52.40	50.18	44.97	34.55	46.03	56.21	46.03
	Baseball Field	88.57	-	86.24	69.60	85.20	87.85	87.33	86.78	87.00	86.58	85.73	86.81	78.97	85.88	87.93	87.55
Road	Intersection	65.07	-	50.41	51.58	50.91	55.24	54.43	48.05	45.92	46.91	55.57	47.83	44.10	49.76	54.4	57.66
	Roundabout	21.02	-	17.99	12.77	15.01	18.63	18.12	20.78	18.42	17.27	17.92	21.01	13.42	15.48	19.80	16.76
	Bridge	11.94	-	18.82	14.94	11.14	17.27	12.56	1.13	11.59	13.04	18.87	0.62	7.71	6.56	23.85	24.39
Overall mAP		35.16	37.1	32.87	15.96	28.06	33.72	32.43	22.91	26.41	29.75	33.11	33.44	25.90	27.21	37.09	38.26

We directly present the results of TIOE-Det and Manba-YOLO from the works [56], [57].

Compared with the second-best methods, i.e., OR-CNN, our proposed method achieves 1.17-mAP improvement. Compared with other object detection architectures, the proposed method shows superior performance. For example, compared with one-stage detector like RetinaNet, our proposed method achieves 15.36-mAP improvement. Compared with the transformer-based method like RoI-Trans, our proposed method obtains a 4.54-mAP improvement. In addition, compared with the full-convolution-based and FOD methods like FOUS and PCLDNet, our proposed method achieves 10.2- and 5.39-mAP improvement, respectively. Compared with the YOLO-based methods, i.e., TIOE-Det and Manba-YOLO, our proposed method achieves 3.1- and 1.16-mAP improvement, respectively. As there is a substantial similarity of different fine-grained objects in the same coarse-grained category, the outperformance of our proposed method demonstrates that our method has a better capacity for modeling discriminative features.

D. Ablation Study

In this section, we perform ablation studies to demonstrate the effectiveness of the proposed learning method, i.e., the DLL method.

1) Qualitative Analysis: Fig. 6 shows the visualization comparisons of our proposed method and the baseline method. It can be seen that the proposed method achieves superior visual detection results compared with the baseline method. Despite the images in Fig. 6 having complex backgrounds, our method and baseline can accurately localize fine-grained objects. However, the baseline method cannot effectively recognize objects as accurate categories. For instance, as shown in the scene of ocean waves in the first column images of Fig. 6, our proposed method accurately localizes and recognizes the ship as an OilTanker. The baseline method mistakenly recognizes the ship as a ContainerShip. In the scene of dense objects in the second column images of Fig. 6, our proposed method accurately recognizes the four ships as Other Warship. In contrast, the baseline method cannot accurately recognize them.

2) Quantitative Analysis: Due to the baseline method being OR-CNN [42], we can see the performance of the baseline method in Tables I and II. The baseline method achieves mAP of 37.09 and 57.5 on the Fair1M and ShipRSImageNet datasets, respectively. Compared with the baseline method, our method improves the mAP by 1.17 and 1.1 on the Fair1M and ShipRSImageNet datasets, respectively.

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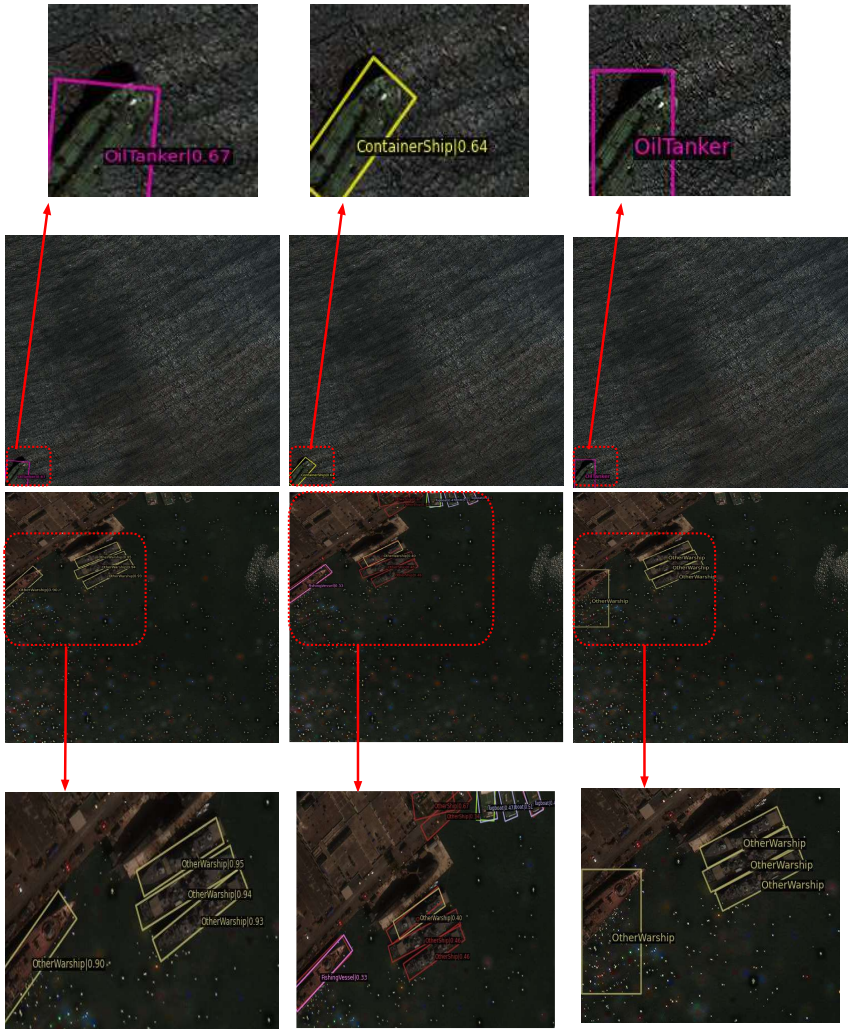


Fig. 6. Qualitative comparisons of the baseline method and our method. GT denotes the ground truth.

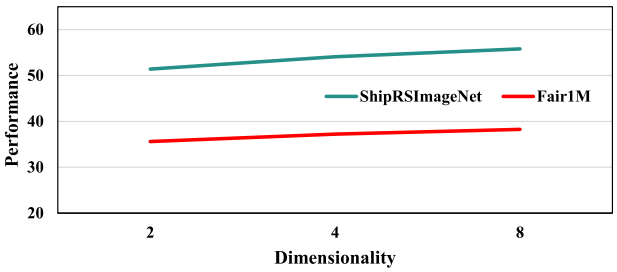


Fig. 7. Performance curve of our proposed method by using different dimensionality parameters.

To evaluate the impact of latent-space dimensionality on performance, we set different dimensionality parameters for comparison. As our proposed method needs additional matrix computation and GPU memory is limited, there are relatively few choices of latent-space dimensionality. Fig. 7 shows the performance curve of our proposed method using different parameters. As shown, the performance achieved by setting a different dimensionality varies in a relatively large range. Higher dimensionality means better performance because the more general correlation is addressed.

3) Universality Analysis: To analyze the universality of the proposed method, we conduct experiments under different method placements. Normally, a two-stage object detection framework consists of three key components, i.e., backbone, RPN-head, and RoI-head. The backbone is used for the extraction of semantic features, and the RPN-head is used for proposal generation. Based on the input image feature and proposal, the RoI-head conducts the final classification and regression. The proposed method is designed for feature modeling. As shown in Fig. 8, we have two options for method placement, i.e.: 1) after the backbone and 2) after the RoI-head. Note that for option 2), we add the DLL after the proposal feature maps and before the classification and regression branch.

In Table III, we show the quantitative results of different method placements. As shown, these two method placements exhibit superiority compared with the baseline method, concluding that the proposed method is relatively universal for different placements. For the reason that method placement option 1) slightly outperforms option 2), we give the following analysis. The backbone is composed of a series of convolution layers, extracting the whole image feature. ROI-head generates proposal features by extracting a series of fixed-size local

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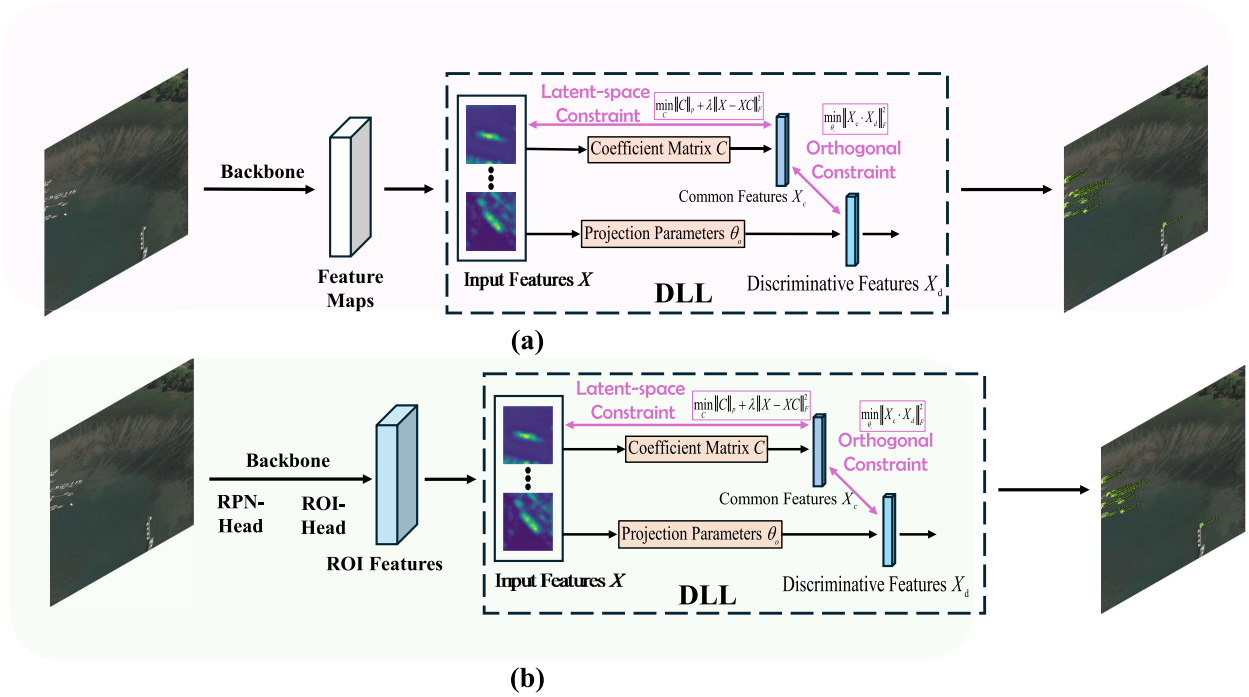


Fig. 8. Visualization of different method placements. (a) DLL is added after the feature maps. (b) DLL is added after the RoI-features.

TABLE III

RESULTS OF DIFFERENT METHOD PLACEMENT EXPERIMENTS ON THE SHIPRSIMAGENET DATASET. DLL WITH FEATURE MAPS DENOTES THE MODEL THAT ADDS THE PROPOSED METHOD AFTER THE BACKBONE, MODELING DISCRIMINATIVE FEATURES BETWEEN IMAGE FEATURES. DLL WITH ROI FEATURES DENOTES THE MODEL THAT ADDS THE PROPOSED METHOD AFTER THE ROI-EXTRACTOR, MODELING DISCRIMINATIVE FEATURES BETWEEN ROI FEATURES. THE BEST RESULTS ARE GIVEN IN BOLD

	Baseline	DLL with feature maps	DLL with RoI features
mAP	57.5	58.6	**57.9**

features based on the proposal position. This leads to the difference in semantic feature continuity between these two method placements.

E. Effectiveness Analysis

In practice, the feature distance between fine-grained objects is much smaller than the feature distance between coarse-grained objects. This gap poses challenges for the model to detect. In this situation, modeling discriminative features with higher distances is one of the critical factors for improving fine-grained detection performance.

As discussed and proven in Section III, the key idea of our proposed method is to extract common underlying features between different category objects and model the irrelevant feature of the input features to common features. Thus, the correlation between features of different fine-grained categories should be affected when the proposed method processes input features. In this section, we conduct deep analysis experiments to further demonstrate the effectiveness of the proposed method.

TABLE IV

FEATURE DISTANCE COMPARISON BETWEEN DIFFERENT COMPONENTS, I.E., INPUT FEATURE, LR FEATURE, AND DISCRIMINATIVE ORTHOGONAL DECOUPLING (DOD). "WITH LR" REPRESENTS FEATURES GENERATED BY LR. "WITH DOD" DENOTES FEATURES GENERATED BY DOD. THE DISTANCE IS CALCULATED BETWEEN TWO BATCH IMAGES RANDOMLY SELECTED FROM THE SHIPRSIMAGENET DATASET USING THE MEAN OF EUCLIDEAN DISTANCE

Input Classes	Input Feature	with LR	with DOD
Motorboat/ContainerShip	0.0272	0.0002462	0.2367
ContainerShip/OilTanker	0.0272	0.0002486	0.2375
OilTanker/Hovercraft	0.0271	0.0002348	0.2381
Hovercraft/Cargo	0.0271	0.0002355	0.2381
Cargo/Ferry	0.0273	0.0002399	0.2385
Ferry/Landing	0.0273	0.0002521	0.2379
Landing/FishingVessel	0.0274	0.0002478	0.2376
FishingVessel/Destroyer	0.0270	0.0002466	0.2371
Multiclasses/Multiclasses	0.0277	0.0002461	0.2384
Multiclasses/Destroyer	0.0277	0.0002461	0.2384

Table IV shows the change of feature distance between different components of our model, i.e., input, LR, and DOD. As shown, for the images with a singer object as input (Motorboat/ContainerShip), the distance between the input

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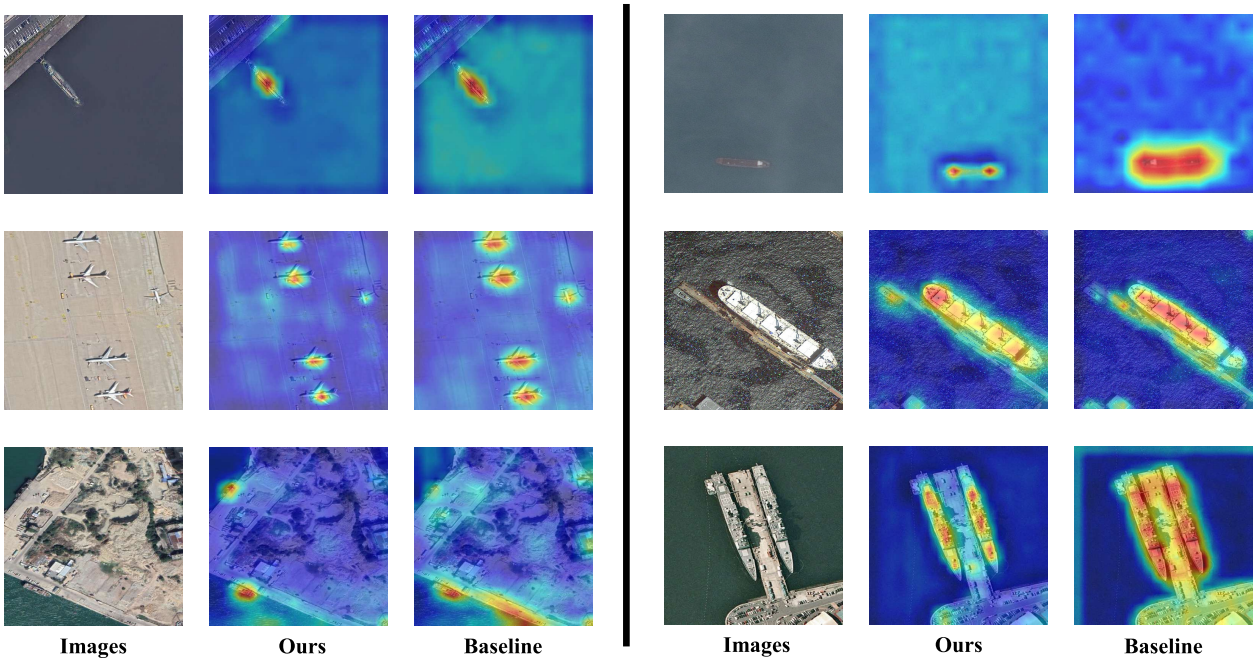


Fig. 9. Grad-CAM visual comparison of baseline and our method. The redder the area, the more the model is sensitive to the area.

features is 0.0272. After the LR, the distance between features is down to 0.0002462. The reason is that we introduce the LR method to extract common underlying features of the input features. As proven in Section III, by minimizing (5), we have the LR that captures shared underlying information between input features. After the DOD, the distance between features is up to 0.2367. The reason is that we introduce the DOD to extract discriminative features. Based on Definition 1 and Theorem 2, we proved that DOD can model irrelevant input features to common features. For the multiobjects as input (multiclasses/multiclasses and multiclasses/destroyer), we can get a similar change in feature distance. This demonstrates that our method can effectively model discriminative features.

Fig. 9 visualizes the Grad-CAM [59] images generated by the baseline and our method, respectively. Grad-CAM is a visual explanation from deep networks via gradient-based localization. It describes the localization of the model that is most sensitive to the input image.

As shown in the second row of Fig. 9(a), compared with the Grad-CAM images generated by the baseline method, the sensitive region of Grad-CAM images generated by our method is more local. As shown in the first row of Fig. 9(b), the baseline method can localize the ship object while failing to finely distinguish the features of different regions. Our method not only localizes the ship object but also can localize discriminative features of the bow and stern of the ship. This proves that our method can effectively improve the sensitivity of the baseline method to local discriminative regions.

F. Visualization of Training Behavior

The previous studies [8], [9] have found that different convolutional feature channels correspond to a different local responsiveness. This local responsiveness represents the degree to which the model is sensitive to local features and hence plays a key role in FOD.

The proposed method tries to model discriminative features between different classes of fine-grained objects using a specifically designed optimization paradigm. As an ideal case, the proposed method should make the model more focused on discriminative local features.

To qualitatively verify the effectiveness of the proposed method to local features, we illustrate the feature maps of our model during the training process in Fig. 10. We can see from (a1) that the feature maps contain basic texture features of the ship object. This demonstrates that the model trained for one epoch can extract coarse texture information of the ship object. As the training process continues, it can be seen that the texture attention of the feature maps becomes more detailed. For Fig. 10(a), it can be seen from Fig. 10(a5) that the highest responsiveness of Fig. 10(a5) is the center of the ship object. Compared with Fig. 10(a1), Fig. 10(a5) shows lower background responsiveness and higher local region responsiveness. Fig. 10(b) shows the feature map of the change process of another scene image as input. It can be seen that the process of feature map change is similar to image Fig. 10(a). This demonstrates that our proposed method can effectively improve the sensitivity of the baseline model to discriminative local features.

G. Analysis of Failure Cases

The above experimental analysis demonstrates the effectiveness of the proposed method. In this section, we analyze the failure cases. As shown in Fig. 11, it can be seen that our proposed method still has limitations. Specifically, for the first and second row images, our proposed method mistakenly detects the objects, i.e., mistakenly classifies the tower as ContainerShip in the first row images and the tail wave as Landing. The main reason may be that the proposed method cannot learn the discriminative information from the background without ground truths during optimization. This results in suboptimal

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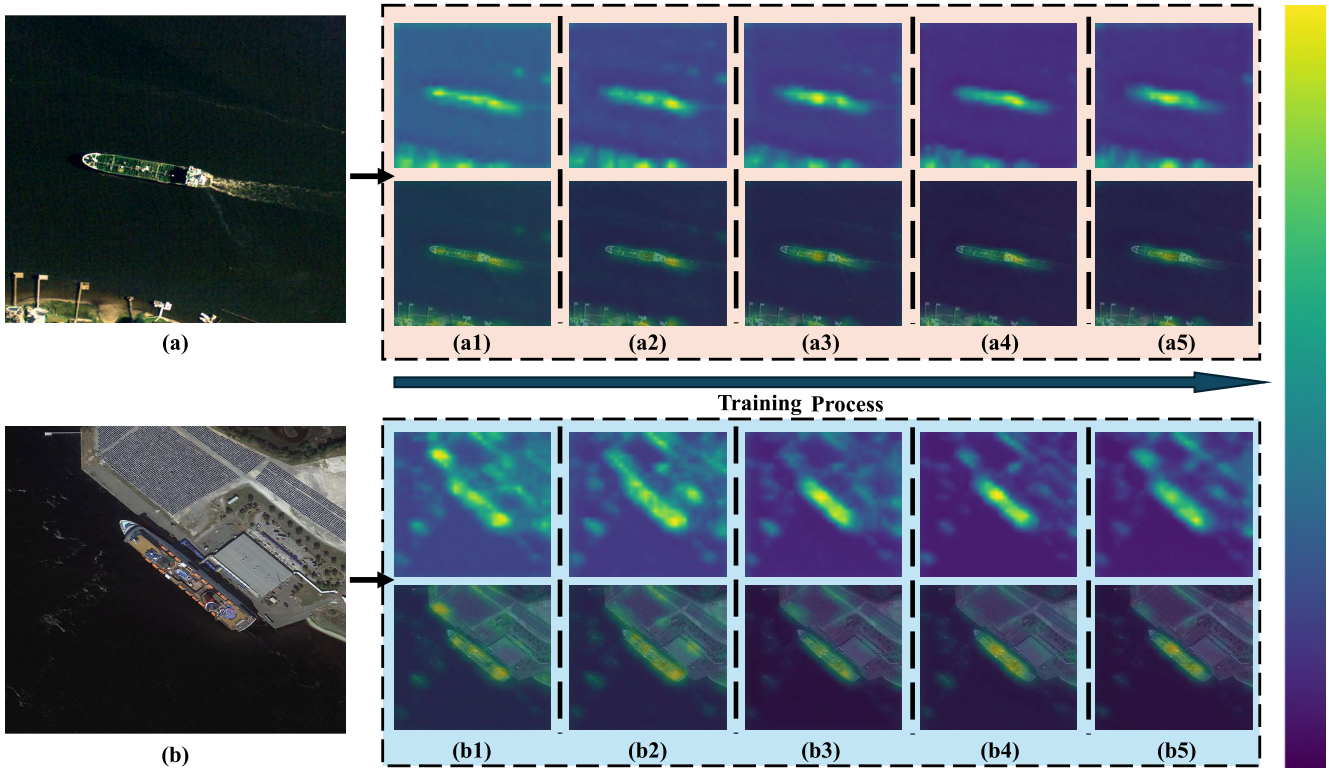


Fig. 10. Comparison of feature maps of different training stages. (a) and (b) Input images. (a1)–(a5) and (b1)–(b5) Feature maps. In the black dashed box, the images from left to right represent the feature maps from the beginning of training to the end of training. The first row denotes feature maps, and the second row illustrates the feature maps mapped on the input images.

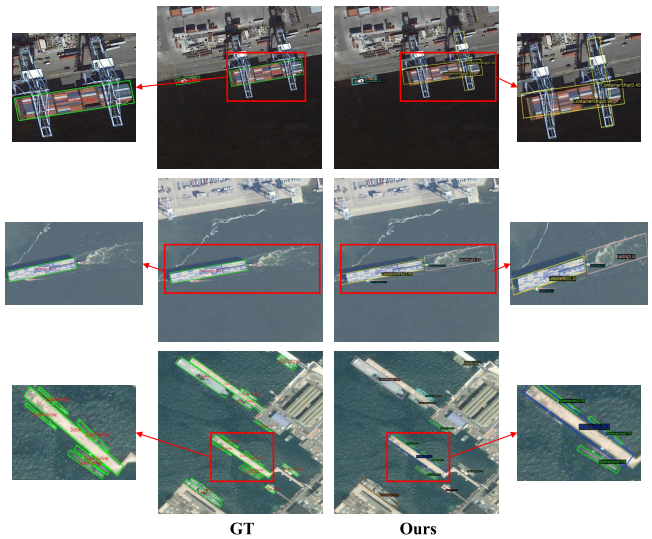


Fig. 11. Failure cases.

background suppression performance improvement. For the third row images with small and dense objects, it can be seen that the missed submarine object is in the water, which causes an unclear feature case. Our proposed method misses a submarine object. The main reason may be that unclear feature cases hinder our proposed method of extracting discriminative features. This discussion of failure cases provides a promising direction for our future work.

VI. Conclusion

In this article, we propose a novel DLL method for FOD in remote sensing images. Existing methods have not considered global discriminative feature modeling and the curse of dimensionality (CoD), resulting in suboptimal discriminative feature extraction. To overcome this issue, the proposed method is designed as a subspace-based method that recovers low-dimensional structures from high-dimensional feature space. Specifically, we first use LR to extract common underlying features between different fine-grained objects. Then, DOD is used to extract the irrelevant features related to common underlying features from input features. In this way, we not only achieve dimensional reduction but also can effectively highlight discriminative features and hence alleviate the CoD. Our method exhibits superior performance compared with other state-of-the-art methods. In the Fair1M and ShipRSImagesNet datasets, our DLL achieves mAP as high as about 38.26 and 58.6, respectively. In addition, the theoretical and experimental analysis demonstrates the effectiveness of the proposed method.

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